

1. Mọi con đường về 0

```
#include<bits/stdc++.h>
using namespace std;
void induong(int s,int f,int *d)
{
    stack<int> S;
    while(f!=s)
    {
        S.push(f);
        f=d[f];
    }
    cout<<s;
    while(S.size())
    {
        cout<<"->"<<S.top();
        S.pop();
    }
}
void Path(int s,int f,int *d)
{
    if(s==f) cout<<s;
    else {Path(s,d[f],d); cout<<"->"<<f;}
}
void dfs(int s,int f) //di sau bao nhieu buoc
{
    stack<int> S;
    int d[s+5]={}; //mang cha
    S.push(s);
    d[s]=-1;
    while(S.size()>0) //while(not S.empty())
    {
        int u=S.top(); S.pop();
        for(int a=1;a*u<=u;a++)
            if(u%a==0)
            {
                int v=(a-1)*(u/a+1);
                if(d[v]==0 && v>=f) {d[v]=u; S.push(v);}
            }
    }
}
```

```

        if(d[f]>0)
        {
            cout<<"co duong di : ";
            Path(s,f,d);//induong(s,f,d);
            return;
        }
    }
    cout<<"khong co duong di";
}
int main()
{
    int s,f;
    cin>>s>>f;
    dfs(s,f);
}

```

2. Bài toán đong nước

```

typedef pair<int,int> Water;
map<Water,Water> M;
void Path(Water x)
{
    if(x.first==0 && x.second==0) cout<<"(0,0)";
    else {Path(M[x]);cout<<"->("<<x.first<<","<<x.second<<");}
}
void BFS(int n,int m,int k)
{
    queue<Water> Q;
    Q.push({0,0});
    M[{0,0}]=-1,-1};
    while(Q.size())
    {
        Water u=Q.front();
        Q.pop();
        int x=u.first,y=u.second,t=x+y;
        Water Next[]={ {0,y},{n,y},{x,0},{x,m},{max(0,t-m),min(t,m)},{min(t,n),max(0,t-n)}};
        for(auto v:Next)
            if(M.find(v)==M.end())

```

```

        {
            M[v]=u;
            Q.push(v);
            if(v.first==k or v.second==k) {Path(v);return;}
        }
    }
    cout<<"\nKhong dong duoc nuoc";
}
int main()
{
    int n,m,k;
    cin>>n>>m>>k;
    BFS(n,m,k);
}

```

3. Phân vùng ảnh

```

#include<bits/stdc++.h>
using namespace std;
struct anh
{
    int n,m,a[100][100],k=0,d[1000]={}; //k- so vùng, d[i] so so 0 o vùng thu i
    void nhap()
    {
        cin>>n>>m;
        for(int i=1;i<=n;i++)
            for(int j=1;j<=m;j++) cin>>a[i][j];
        for(int i=0;i<=n+1;i++) a[i][0]=a[i][m+1]=1;
        for(int j=0;j<=m+1;j++) a[0][j]=a[n+1][j]=1;
    }
    void dfs(int u,int v)
    {
        d[k]++;
        a[u][v]=1;
        for(int i=-1;i<=1;i++)
            for(int j=-1;j<=1;j++) if(a[u+i][v+j]==0) dfs(u+i,v+j);
    }
    void sol()
    {

```

```

        nhap();
        for(int i=1;i<=n;i++)
        for(int j=1;j<=m;j++)
        if(a[i][j]==0) {k++;dfs(i,j);}
        cout<<"\nSo vung " <<k<<"\n";
        sort(d+1,d+k+1);
        for(int i=1;i<=k;i++) cout<<d[i]<<" ";
    }
};
int main()
{
    anh A;
    freopen("anh.txt","r",stdin);
    A.sol();
}

```

4. Cách ly Covid

```

#include<bits/stdc++.h>
using namespace std;
int main()
{
    int n,m,d[100000],F[10000]={},x,y;
    vector<int> A[10000]; //A[1] chua cac phan tu tiep xuc voi 1
    queue<int> Q;
    cin>>n>>m;
    while(m--)
    {
        cin>>x>>y;
        A[x].push_back(y);
        A[y].push_back(x);
    }
    fill(d,d+n+1,-1); //for(int i=0;i<=n;i++) d[i]=-1;
    cin>>F[0];
    for(int i=1;i<=F[0];i++)
    {
        cin>>x;
        Q.push(x);
        d[x]=0;
    }
}

```

```

    }
    while(Q.size())
    {
        int u=Q.front(); Q.pop();
        for(auto v:A[u])
            if(d[v]==-1)
            {
                d[v]=d[u]+1;
                Q.push(v);
                F[d[v]]++;
            }
    }
    for(int i=0;F[i]!=0;i++)
    {
        cout<<"\nF"<<i<<" : "<<F[i]<<" : ";
        for(int j=1;j<=n;j++) if(d[j]==i) cout<<j<<" ";
    }
}

```

5. Buôn dưa lê

```

#include<bits/stdc++.h>
using namespace std;
int main()
{
    int n,m,k,x,s=0,t;
    queue<int> Q;
    cin>>n>>k>>m;
    for(int i=1;i<=n+k;i++)
    {
        if(i<=n) cin>>x; else x=0;
        Q.push(x);
        if(Q.size()>k) Q.pop();
        t=0;
        while(Q.size() && Q.front()+t<=m) {t+=Q.front(); Q.pop();}
        if(Q.size()) {Q.front()-=m-t;t=m;}
        s+=t;
    }
    cout<<s;}

```

6. Lập lịch

```
#include<bits/stdc++.h>
using namespace std;
#define bd second
#define kt first
int main()
{
    int n,t=-INT_MAX,s=0;
    cin>>n;
    vector<pair<int,int>> A(n);
    for(auto &x:A) cin>>x.bd>>x.kt;
    sort(A.begin(),A.end());
    for(auto x:A)
        if(x.bd>=t)
        {
            s++;
            t=x.kt;
        }
    cout<<s;
}
```

7. Trình thám

```
#include<bits/stdc++.h>
using namespace std;
int main()
{
    int n,k,x;
    priority_queue< pair<int,int> > Q;
    cin>>n>>k;
    for(int i=1;i<=n;i++)
    {
        cin>>x;
        while(Q.size() && i-Q.top().second>=k) Q.pop();
        Q.push({x,i});
        if (i>=k) cout<<Q.top().first<<" ";
    }
}
```

8. Nối thanh kim loại

```

#include<bits/stdc++.h>
using namespace std;
int main()
{
    int n,x,res=0;
    priority_queue<int,vector<int>,greater<int>>Q;
    cin>>n;
    while(n--){cin>>x;Q.push(x);}
    while(Q.size()>1)
    {
        x=Q.top(); Q.pop();
        x+=Q.top(); Q.pop();
        Q.push(x);
        res+=x;
    }
    cout<<res;
}

```

9. Phần tử trung vị

```

#include<bits/stdc++.h>
using namespace std;
int main()
{
    priority_queue<int> L;
    priority_queue<int,vector<int>, greater<int> > R;
    int n,x;
    cin>>n;
    for(int i=1;i<=n;i++)
    {
        cin>>x;
        i%2?L.push(x):R.push(x);
        if (i>1 and L.top()>R.top())
        {

```

```

        int u=L.top();L.pop();

        int v=R.top();R.pop();

        L.push(v);

        R.push(u);

    }

    cout<<L.top()<<" ";

}

```

10. Quân hậu

```

#include<bits/stdc++.h>
using namespace std;
map<int,bool> C,A,B; //C-cot, A-duong cheo chinh, B - duong cheo phu
int dem=0;
void TRY(int *x,int k,int n) //gs da co x1...xk
{
    if(k==n)
    {
        cout<<"\n"<<++dem<<" : ";
        for(int i=1;i<=k;i++) cout<<"("<<i<<","<<x[i]<<") ";
        return;
    }
    for(int t=1;t<=n;t++)
    if(C[t]==0 && A[k+1-t]==0 && B[k+1+t]==0 )
    {
        x[k+1]=t;
        C[t]= A[k+1-t]= B[k+1+t]=1;
        TRY(x,k+1,n);
        C[t]= A[k+1-t]= B[k+1+t]=0;
    }
}
int main()
{
    int n=4,x[1000];
    cin>>n;
    // A[0]=1; B[n+1]=1;

```



```

    TRY(x,0,n);
}

```

11. Đòi tiền

```

#include<bits/stdc++.h>
using namespace std;
int main()
{
    int n,a,C[10005]={},M,t;
    cin>>n;
    for(int j=1;j<=1e4;j++) C[j]=1e9;
    for(int i=1;i<=n;i++)
    {
        cin>>a;
        for(int j=a;j<=1e4;j++) C[j]=min(C[j],1+C[j-a]);
    }
    cin>>t;
    while(t--) {cin>>M; cout<<C[M]<<"\n";}
}

```

12. Sắp xếp balo

```

#include<bits/stdc++.h>
using namespace std;
int a[100],b[100],C[100][1005]={},n,M;
void Trace(int n,int M)
{
    if(C[n][M]>0)
    {
        while(C[n][M]==C[n-1][M]) n--;
        Trace(n-1,M-a[n]);
        cout<<"Chon do vat kt "<<a[n]<<" gt "<<b[n]<<"\n";
    }
}
int main()
{
    cin>>n>>M;
    for(int i=1;i<=n;i++) cin>>a[i]>>b[i];
    for(int i=1;i<=n;i++)
    for(int j=1;j<=M;j++) C[i][j]=a[i]>j?C[i-1][j]:max(C[i-1][j],b[i]+C[i-1][j-a[i]]);
}

```

```

        cout<<"\nTong gt lon nhat "<<C[n][M]<<"\n";
        Trace(n,M);
    }

```

13. Cắt hình vuông ít nhất

```

#include<bits/stdc++.h>
using namespace std;
int C[100][100],P[100][100];
void trace(int n,int m,string p="\n")
{
    if(n!=m) {if(P[n][m]>0) trace(P[n][m],m,p+"\t"); else trace(n,-P[n][m],p+"\t");}
    cout<<p<<n<<"*"<<m;
    if(n!=m) {if(P[n][m]>0) trace(n-P[n][m],m,p+"\t"); else
    trace(n,m+P[n][m],p+"\t");}
}
int main()
{
    int n,m;
    cin>>n>>m;
    for(int i=1;i<=n;i++) {C[i][1]=i; P[i][1]=1;}
    for(int j=1;j<=m;j++) {C[1][j]=j; P[1][j]=-1;}

    for(int i=2;i<=n;i++)
        for(int j=2;j<=m;j++)
            if(i==j) C[i][j]=1;
            else
            {
                C[i][j]=i*j;
                for(int k=1;k<=i/2;k++) if(C[i][j]>C[k][j]+C[i-k][j])
                {C[i][j]=C[k][j]+C[i-k][j]; P[i][j]=k;}
                for(int k=1;k<=j/2;k++) if(C[i][j]>C[i][k]+C[i][j-k])
                {C[i][j]=C[i][k]+C[i][j-k]; P[i][j]=-k;}
            }
    cout<<"\nsố hình vuông ít nhất "<<C[n][m]<<"\n\n";
    trace(n,m);
}

```

14. Xâu con chung dài nhất

```

#include<bits/stdc++.h>

```

```

using namespace std;
string x,y;
int C[105][105]={};
void trace(int n,int m)
{
    if(C[n][m]>0)
    {
        while(C[n][m]==C[n-1][m]) n--;
        while(C[n][m]==C[n][m-1]) m--;
        trace(n-1,m-1);
        cout<<x[n];
    }
}
int main()
{
    int n,m;
    cin>>x; n=x.size(); x="$"+x;
    cin>>y; m=y.size(); y="$"+y;
    for(int i=1;i<=n;i++)
    for(int j=1;j<=m;j++)
        if(x[i]==y[j]) C[i][j]=1+C[i-1][j-1];
        else C[i][j]=max(C[i-1][j],C[i][j-1]);
    cout<<"\nDo dai xccd n "<<C[n][m]<<"\n";
    trace(n,m);
}

```

15. Truy vấn max

```

#include<bits/stdc++.h>
using namespace std;
struct node
{
    int a,b,elem; //quan ly elem=max(a[a]....a[b-1])
    node *left,*right;
    node(int u,int v)
    {
        a=u; b=v; elem=-INT_MAX;
        if(u+1==v) left=right=NULL;
        else {left = new node(a,(a+b)/2); right=new node((a+b)/2,b);}
    }
}

```

```

    }
};

void update(node *H,int i,int x)
{
    if(H->elem<x) H->elem=x;
    if(H->left) update(i<H->left->b?H->left:H->right,i,x);
}

int get(node *H,int L,int R)
{
    if(H->a==L && H->b==R) return H->elem;
    if(R<=H->left->b) return get(H->left,L,R);
    if(L>=H->right->a) return get(H->right,L,R);
    return max(get(H->left,L,H->left->b),get(H->right,H->right->a,R));
}

int main()
{
    int n,x,L,R,m;
    cin>>n>>m;
    node *root=new node(1,n+1);
    for(int i=1;i<=n;i++) {cin>>x; update(root,i,x);}
    while(m--)
    {
        cin>>L>>R;
        cout<<get(root,L,R)<<"\n";
    }
}

```

16. Đếm số nghịch thế

```

#include<bits/stdc++.h>
using namespace std;
int A[400005]={};
long long res=0;
void update(int k,int a,int b,int x) //A[k] quan ly [a,b) cap nhat x
{
    A[k]++;
    if(a+1<b)
    {
        if(x<(a+b)/2) {res+=A[2*k+2]; update(2*k+1,a,(a+b)/2,x);}
    }
}

```

```
        else update(2*k+2,(a+b)/2,b,x);
    }
}
int main()
{
    int n,x;
    cin>>n;
    for(int i=1;i<=n;i++)
    {
        cin>>x;
        update(0,1,n+1,x);
    }
    cout<<res;
}
```